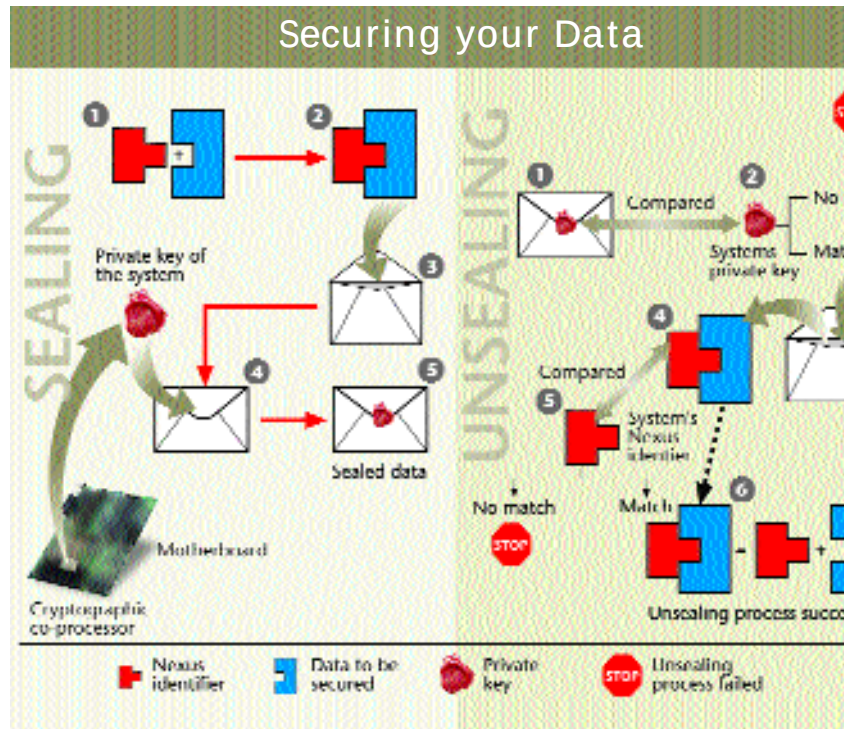


Think information

Using the Nexus and the supporting hardware, a Palladium system is cryptographically verifiable to the user and to other computers or services. The system can ascertain that other computers are trustworthy before engaging them in a transaction involving sensitive data. This is termed as attestation and it forms a cornerstone of successful security and privacy measures. Take the instance of a bank operation. Banking software has to be highly trusted and moreover, it must be ensured that the computer dealing with it does indeed belong to you and not to some malicious party masquerading under your identity. You hardware should thus trust the banking software and vice-versa. This virtual handshake to exchange and verify credentials is made possible by attestation.

Once attestation is done, exchange of data takes place. It is important that data is only accessible to applications that have been identified as trusted. Once your personal data is transmitted to the bank, you could so arrange the transaction that only particular software can read your information. Such control is made possible by a mechanism called sealed storage.

Sealed storage is done by encryption or decryption techniques using the crypto-



graphic co-processor. To seal a piece of data, it is appended with the hash of the system's Nexus identifier (a 160-bit number). It is then encrypted with a private key—a secret key unique to each system. When a sealed data is given to a Palladium system, the co-processor attempts to decrypt it using its platform-specific private key. If successful, the hashed Nexus identifier is extracted and decrypted to determine which Nexus is permitted to receive the data. If the Nexus that currently resides on the computer is the same as the one that was present while the data was sealed, the data is unsealed and read. Two things emerge from this:

1. You cannot take a sealed file and transfer it to another machine to be unsealed there since the platform-specific key will change and the process will fail at the first step. Such a measure is essential if you wish to store sensitive data on a non-trusted hard disk or over a network. Since this is an inconvenience to most users who demand transportability of their data, Microsoft has defined wrappers, which are software extensions that will implement one of three migration policies: (a) migration is entirely prevented by default setting, (b) migration is permitted upon entry of a valid password, (c) migration is allowed with the consent of a third party, for instance, permission is required while downloading an MP3 file from a music company.
2. The same Nexus must be present while sealing and unsealing the data. If you have changed the micro-kernel after

encrypting a file, you will have to go to the older one to read the file.

Thus, your applications can decrypt sealed data if they run on the hardware and under an identical so environment (as dictated by the N under which the data was sealed. ensures that under a trusted Nex application can decrypt and proce data but cannot subvert a TA's pol you're not running under a trusted I you could run software that under the policy set by a TA, but the appli will no longer be able to decrypt the data since the Nexus has changed (se diagram, 'Securing your Data').

Think implications

Palladium does not require user information for authentication purposes; instead it secures the hardware and software environment. The Nexus identifier is unique to a system, since it is likely that the majority of people will use the same I. The only identifier unique to your system will thus belong to the cryptographic co-processor. Although Palladium discourages the transmission of any private information, Microsoft has defined the concept of identity servers. These are third-party services that will grant you a unique identity credential using Palladium, and they may not reveal personal information.

The implications of using Palladium are many and not all scenarios m

A Nice Thought

Palladium is widely touted as a system that will put an end to virus attacks and spam. Can it hope to achieve those lofty goals?

How does Palladium hope to stop viruses?

Trusted code runs in memory that is physically isolated, protected and inaccessible to the rest of the system, making it inherently impervious to viruses.

Most viruses exploit the way software applications use scripting. A virus can thus infect a trusted application and use it maliciously without infecting memory space. You will still need antivirus software.

How does Palladium hope to stop spam?

A Palladium enabled e-mail client will only download mail from 'trusted' machines and servers, thus spammers are automatically blocked.

Spamming is a very lucrative business. To get around the security measure imposed by Palladium, all a spammer had to do is use a trusted machine to do the dirty work.